Hi An,
As discussed, please go thru the following points which need to be updated in the code I shared:

1. EEPROM Function:
Identify the first bootup and when captured as First Boot Up, ask the user to scan Master Card (RFID Card), and Production Card (RFID Card).
Remove the requirement and memory space for QUALITY CARD and MAINTENANCE CARD, from EEPROM and SETTINGS MENU (DELETE/MODIFY)

2. Timer resolution to be made to 1ms. (ADD/MODIFY)

3. Lt will be set as per the attached sheet. (AVAILABLE, MODIFY LT AS PER SHEET)

4. During Batch counting , in HT error condition-reverse counting doesn't works which is okay but in LT error it counts the reverse minus which is not ok. (AVAILABLE, NEED TO VERIFY AND MODIFY)

5. Inputs (AVAILABLE, NEED TO VERIFY AND MODIFY):
Before sending Fout = LOW, two conditions need to be checked: 1. part presence sensor signal-- it is a continuous signal. The microcontroller will wait for the signal and send Fout LOW only when this has been detected.
2. Clamp button: this is a button press. The microcontroller will wait for the button press and send Fout LOW only when this has been detected.

These input signals are being received from PCF8574AT IC (this IC has been used for GPIO expansion).

In the settings menu, there is a provision to bypass these signals. CLAMP and PROX if selected as OFF in settings, Fout signal will not look for Part presence sensor and Clamp button.
They both have individual settings. Meaning user can choose to by-pass either or both of them.

6. SOS functionality (TIHS NEEDS TO BE ADDED IN THE CODE): When this button is pressed, AOK signal is generated and SC Count is reset to 0 (zero). Since there is no hardware button available for this now, we need to read two different buttons (please select as per convenience) and use both button press together as SOS button press.